

Fixed-Ro equal-area theory on the equatorial β plane, for the SS09 shallow-water model

pengcheng-ss09 suite, Tier 0 (A0 memo)

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Beta-plane translations of the closed forms in the fixed-Ro manuscript (`papers/fixed-ro-eq-area_ms-draft-v1.pdf`, “ms” below), derived against this model’s exact equations. Every result carries a basis tag; “pending run” tags get upgraded as the suite’s runs confirm them. Section 8 lists the falsifying checks and their status, and the run record follows it.

1 Setup and notation

Steady state, hemispherically symmetric forcing ($y_0 = 0$), northern hemisphere ($y > 0$). The model’s steady equations (`SCIENCE.md` section 2), with $T = \theta/1.6$ and the v equation’s factor of two multiplying only the time tendency (so it drops out of every steady balance):

$$v \left(\beta y - \frac{\partial u}{\partial y} \right) = \mathcal{H}(\theta_E - \theta) u \frac{\partial v}{\partial y} + \epsilon_u u + v_d \mathcal{H}(u) \operatorname{sgn}(y) \frac{\partial u}{\partial y}, \quad (1)$$

$$\beta y u = -\frac{gH}{T_0} \frac{\partial T}{\partial y} + k_v \frac{\partial^2 v}{\partial y^2}, \quad (2)$$

$$\frac{\delta \Delta_z}{H} \frac{\partial v}{\partial y} = \frac{\theta_E - \theta}{\tau}. \quad (3)$$

Forcing (SS09 parabolic profile, the theory-matched choice): $\theta_E = \theta_{00} - \Delta_y (y/y_1)^2$ for $|y| < y_1$. In temperature units,

$$T_E(y) = T_E(0) - b y^2, \quad b \equiv \frac{\Delta_T}{y_1^2}, \quad \Delta_T \equiv \frac{\Delta_y}{1.6}, \quad (4)$$

with $T_0 = 300$ K. The small-angle limit of the ms is exact here: the β plane has no $\cos \varphi$ factors, so no truncation enters any of the following.

2 The one-line mapping

Every ms result carries over with the substitutions

$$\operatorname{Ro}_{\text{th}} \rightarrow R_\beta \equiv \frac{4gH\Delta_T}{T_0\beta^2 y_1^4}, \quad \Delta_h \theta_0 \rightarrow \Delta_T, \quad a\varphi \rightarrow y, \quad \varphi_{\text{Ro}} \rightarrow Y_{\text{Ro}}/y_1. \quad (5)$$

[**derived**] R_β follows from the sphere definition $\operatorname{Ro}_{\text{th}} = gH\Delta_h/(\Omega a)^2$ with $\Omega = \beta a/2$ and $\Delta_h = \Delta_T a^2/(T_0 y_1^2)$ (matching the forcing curvature at the equator); the planetary radius cancels, as it must on a β plane. At this repo’s defaults ($g = 9.81 \text{ m s}^{-2}$, $H = 16 \text{ km}$, $\Delta_y = 50 \text{ K}$ so $\Delta_T = 31.25 \text{ K}$, $T_0 = 300 \text{ K}$, $\beta = 2 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$, $y_1 = 9439 \text{ km}$):

$R_\beta = 0.0206.$

(6)

3 Wind and temperature

[derived: pending run confirmation] Uniform $\text{Ro} = (\partial_y u)/(\beta y)$ integrates to $u_{\text{Ro}}(y) = \text{Ro} \beta y^2/2$ (ascent at $y = 0$). Gradient balance (neglecting k_v) gives $\partial_y T = -(T_0/gH) \text{Ro} \beta^2 y^3/2$, so

$$T_{\text{Ro}}(y) = T_{\text{Ro}}(0) - c y^4, \quad c \equiv \frac{\text{Ro} T_0 \beta^2}{8gH}. \quad (7)$$

4 Equal-area edge and equatorial temperature

[derived: scalar anchors confirmed at the 1–8% level by runs 1a–1b below; at profile level the inner half of the cell agrees at the 10% level and the terminus misfit is ~ 35 –40% of the depression, run-record addendum] With $D(y) = T_{\text{Ro}}(y) - T_E(y)$, imposing $D(Y) = 0$ and $\int_0^Y D dy = 0$ (measure dy ; the Newtonian relaxation has uniform τ) gives $\frac{2}{3}bY^2 = \frac{4}{5}cY^4$, i.e. $Y^2 = \frac{5}{6}b/c$, which in Section-2 variables is

$$Y_{\text{Ro}} = y_1 \left(\frac{5R_\beta}{3\text{Ro}} \right)^{1/2} \quad [\text{ms Eq. 13 analog}], \quad (8)$$

$$T_E(0) - T_{\text{Ro}}(0) = \frac{5}{18} \frac{R_\beta \Delta T}{\text{Ro}} \quad [\text{ms Eq. 15 analog}]. \quad (9)$$

Defaults, $\text{Ro} = 1$: $Y_{\text{Ro}} = 1.75 \times 10^6$ m and equatorial depression 0.18 K (in T ; multiply by 1.6 for θ units). At $\text{Ro} = 0.15$ the depression is 1.2 K (T) / 1.9 K (θ).

Consistency note: the analogous derivation for the $\sin^2$ profile needs the numerical equal-area solver (puffins `equal_area_bouss` with `Ro`-scaled `del_h_over_ro`, verified in puffins PR #46). Near the equator $\sin^2(\pi y/(2y_1)) = (\pi y/(2y_1))^2 + O(y^4)$, so $\sin^2$ behaves like a parabola with curvature enhanced by $(\pi/2)^2 = 2.47$; Section 7's postdiction check uses exactly this factor.

5 Meridional wind, heat flux, momentum flux

[derived: v_{max} magnitude and location confirmed at the 1–4% level by runs 1a–1b below; at profile level the inner flank agrees at the 5% level and the terminus flank misfit is $\sim 45\%$ of v_{max} , run-record addendum] The steady thermodynamic balance (3) integrates to the model's v profile directly, using $\theta_E - \theta = 1.6 (T_E - T) = -1.6 D$:

$$v(y) = -\frac{1.6H}{\delta \Delta_z \tau} \int_0^y D dy' = \frac{1.6H}{\delta \Delta_z \tau} \frac{5}{18} \frac{R_\beta \Delta T}{\text{Ro}} Y_{\text{Ro}} (x - 2x^3 + x^5), \quad x \equiv \frac{y}{Y_{\text{Ro}}}. \quad (10)$$

The bracket is exactly the ms Eq. (16) shape; it peaks at $x = 5^{-1/2} = 0.447$ with value 0.286, and $v_{\text{max}} \propto \text{Ro}^{-3/2}$ (ms Eq. 20 analog). Defaults, $\text{Ro} = 1$: $v_{\text{max}} = 3.0 \times 10^{-3} \text{ m s}^{-1}$ at $y = 0.78 \times 10^6$ m.

Column fluxes, using the two-layer reading (upper slab of depth δ with velocity $+v$ and potential temperature θ_{top} , return slab of depth δ with $-v$ and θ_{bot} , $\theta_{\text{top}} - \theta_{\text{bot}} = \Delta_z$; return-flow u negligible):

$$\text{heat flux: } \int v \theta dz = \delta \Delta_z v(y) \quad [\text{ms Eq. 16}], \quad (11)$$

$$\text{momentum flux: } \int v u dz = \delta v(y) u_{\text{Ro}}(y) \quad [\text{ms Eq. 17}], \quad (12)$$

$$\text{vertical velocity: } w = \delta \frac{\partial v}{\partial y}, \quad w_{\text{max}} \propto \text{Ro}^{-1} \quad [\text{ms sec. 3f}]. \quad (13)$$

The ms’s counterintuitive momentum-flux result carries over: expanding $\delta v u_{\text{Ro}}$, the leading term in y is Ro-independent (the Ro^{-1} strengthening of v cancels the Ro weakening of u in the product) and $\partial(\text{flux})/\partial\text{Ro} < 0$ strictly inside the cell, exactly as verified for the sphere forms in puffs in `test_eq-area.py`.

6 Diagnosed quantities: conventions for the suite

- **Full-profile scorecard (the confirmation standard for every suite run):** the simulated u , T , and v profiles are compared against the zero-free-parameter theory profiles at the diagnosed (regression) cell-mean Ro, reporting $\max|\Delta|$ and its location both over the cell $[0, Y_{\text{Ro}}]$ and over its inner half, each scaled by the field’s theory amplitude (cell-edge u , the equatorial depression, and v_{max} respectively). The temperature comparison is visualized as the equal-area lobes $D(y) = T - T_E$, which removes the ~ 200 K shared background. Implemented in `scripts/fixed_ro_scorecard.py` (one four-panel figure per run; fields are means over the final 30 days). Scalar anchors (depression, v_{max} , edge) are by-products of the same tool, not the confirmation standard: runs 1a–1b passed the scalars at the 1–8% level while the profile misfit at the terminus is ~ 35 –45% of scale (see the profile addendum in the run record).
- **Local Ro(y):** $(\partial_y u)/(\beta y)$, the model’s existing diagnostic (NaN within $|y| < \Delta y_{\text{grid}}$ of the equator).
- **Cell-mean Ro, two definitions, both reported** (paralleling Hill et al. 2025): (i) the area mean of local Ro over $[y_{\text{Ro-max}}, Y_{\text{edge}}]$; (ii) the best-fit Ro regressing $u(y)$ against $\beta y^2/2$ over the same interval. The near-ascent region is excluded because $\text{Ro} \rightarrow 0$ at the equator whenever $v_d > 0$ (Zhang et al. 2025, $u \sim y^3$).
- **Cell edge, three definitions, all reported:** the v -threshold edge ($|v|$ below 10% of its extremum, existing diagnostic; see run 1a for its failure mode at small v_d), the jet latitude (existing spline diagnostic), and the θ -merge latitude (where θ rejoins θ_E within a tolerance, searched poleward of the *jet*; see run 1a). The theory says they coincide; their splitting is itself a result.
- **Momentum-budget shares:** each steady term of (1) evaluated from output; the drag share $\epsilon_u u$ and the vertical-advection share quantify the accuracy of the EMFD–Coriolis balance underlying Zhang et al. (2025) Eq. 24, $\text{Ro} = v/(v + v_d)$.

7 Emergent cell-mean Ro closure

[derived: pending run confirmation] Combining Zhang’s Eq. 24 at the cell-mean level with Section 5’s v magnitude (the cell mean of the bracket over $[0, 1]$ is $1/6$):

$$\bar{\text{Ro}} = \frac{\bar{v}}{\bar{v} + v_d}, \quad \bar{v}(\bar{\text{Ro}}) = \frac{1.6H}{\delta\Delta_z\tau} \frac{5}{18} \frac{R_\beta\Delta_T}{\bar{\text{Ro}}} \frac{Y_{\text{Ro}}(\bar{\text{Ro}})}{6}, \quad (14)$$

a transcendental equation for $\bar{\text{Ro}}(v_d)$ with no simulation input. This is the suite’s analysis G; it extends Zhang et al. (2025) section 4 (their max-Ro scalings $v_d^{-2/5}$, $v_d^{-1/4}$) to the cell-mean variable the ms uses.

Postdiction check [**computed**]: Section 5’s v with the $\sin^2$ profile’s larger near-equator curvature ($(\pi/2)^2 = 2.47$ in b , entering v through $b^2 Y/c$ as a factor ~ 9.6) predicts $v_{\max} \sim 0.03 \text{ m s}^{-1}$ for the published defaults, matching Zhang et al. (2025) Figs. 2–3 magnitudes within $\sim 1.5\times$.

8 Falsifying checks and status

Every run in the checks below also gets the Section 6 full-profile scorecard on u , T , and v ; the tier-specific items are the additional, targeted comparisons.

1. **Full $u/T/v$ profiles, edge, and equatorial depression vs theory at diagnosed Ro (the most-falsifying single check):** SS09-parabolic forcing, $v_d = 0$. Framing correction from the first run: with the default vertical momentum advection on, $v_d = 0$ is *not* the $\text{Ro} = 1$ limit (Zhang et al. 2025: $u \sim \beta y^2/3$ near the ascent, and drag lowers it further); the theory must be evaluated at the run’s diagnosed cell-mean Ro. The $\text{Ro} \rightarrow 1$ anchor requires `--no-vert-advect-u` and reduced drag (run 1b). Status: runs 1a–1b below, scalar level and profile level (addendum).
2. **Edge collapse:** $Y_{\text{edge}}(\text{Ro}/R_\beta)^{1/2}/y_1 = (5/3)^{1/2}$ flat across the v_d , Δ_y , β , and radius ladders. Status: pending Tiers 1–2.
3. **Radius-ladder null:** dimensional Y_{edge} invariant under $(\beta \rightarrow \beta/k, y_1 \rightarrow ky_1, L \rightarrow kL)$. Status: pending Tier 2.
4. **v -profile shape and $\text{Ro}^{-3/2}$ magnitude scaling** (Section 5). Status: shape checked at profile level in runs 1a–1b (addendum); the $\text{Ro}^{-3/2}$ scaling pending Tier 1.
5. **Momentum flux increases as Ro decreases** (Section 5). Status: pending Tier 1.
6. **Emergent $\bar{\text{Ro}}(v_d)$** (Section 7). Status: pending Tier 1 + analysis G.

Run record

Each confirming or refuting run gets an entry: configuration, output path, measured vs predicted.

2026-07-16, check 1a: $v_d = 0$, default physics (vertical advection on)

Config: SS09 parabolic forcing, $v_d = 0$, $n_y = 801$, $\Delta t = 30 \text{ s}$, numba backend, steady-state stop (window 30 d, threshold 10^{-4}), converged day 959. Output: `model_output/fixed_ro_suite/tier0_amc_edge/amc_ss09_ny801_dt30_vd0.nc`. Steadiness caveat [**computed**]: the jet was still drifting $+0.4\%$ per 60 d at stop (drag-timescale tail), so numbers carry roughly a 2% allowance.

Diagnosed: cell-mean Ro 0.469 (area mean) / 0.470 (u regression); local $\text{Ro}(y)$ is *not* uniform: ~ 0.68 at the ascent (the vertical-advection 2/3 of Zhang et al. 2025, drag-reduced), declining through ~ 0.6 mid-cell to ~ 0.2 approaching the jet, i.e. the ms’s linear-Ro motivation in miniature.

Theory at diagnosed $\text{Ro} = 0.470$ vs run [**computed**]:

quantity	predicted	measured	rel. gap
equatorial depression $T_E(0) - T(0)$	0.381 K	0.379 K	0.5%
$v_{\max}$	$9.28 \times 10^{-3} \text{ m/s}$	$9.39 \times 10^{-3} \text{ m/s}$	1.2%
y of $v_{\max}$ ($0.447 Y_{\text{Ro}}$)	$1.14 \times 10^6 \text{ m}$	$1.18 \times 10^6 \text{ m}$	3.5%
cell edge Y_{Ro} (vs jet latitude)	$2.55 \times 10^6 \text{ m}$	$2.36 \times 10^6 \text{ m}$	-7.4%

Sections 4–5 upgraded to **[confirmed at the 1–8% level, single run]** for these observables at this Ro. Additional findings: the u plateau outside the cell is the parabolic forcing’s RCE gradient wind,

$$u_{\text{RCE}} = \frac{2gH\Delta_T}{T_0\beta y_1^2} = 18.35 \text{ m/s}, \quad (15)$$

constant out to y_1 (verified to 3 digits); the v -threshold edge diagnostic is unusable at small v_d because the drag circulation $v = \epsilon_u u_{\text{RCE}}/(\beta y) \sim 2 \times 10^{-3} \text{ m/s}$ exceeds 10% of v_{max} out to $\sim 9.3 \times 10^6 \text{ m}$; the θ -merge diagnostic must search poleward of the jet, not of the v max, to avoid the equal-area lobe crossing.

2026-07-16, check 1b: $v_d = 0$, vertical advection off

Config: as 1a plus `--no-vert-advec-u`; converged day 971. Output: `.../tier0_amc_edge/amc_ss09_ny801_dt30_vd0_novert.nc`.

Removing vertical advection left the cell-mean Ro essentially unchanged (fit 0.473 vs 0.470; area mean 0.536 vs 0.469), because **the Rayleigh drag, at its default $\epsilon_u = 10^{-8} \text{ s}^{-1}$, is the leading-order non-AMC agent for this forcing [computed]**: the steady zonal momentum balance over the cell interior is

$$v \left(\beta y - \frac{\partial u}{\partial y} \right) = \epsilon_u u \quad \text{to within } \sim 5\text{--}10\% \quad (16)$$

(median ratio 1.05 over $0.3 \times 10^6 < y < 2.2 \times 10^6 \text{ m}$). The parabolic forcing’s weak circulation ($v_{\text{max}} \sim 9 \times 10^{-3} \text{ m/s}$) makes $(1 - \text{Ro}) \sim \epsilon_u u / (v \beta y)$ order one. Local $\text{Ro}(y)$ declines monotonically from ~ 1 at the ascent to ~ 0.2 at the jet.

Theory at diagnosed $\text{Ro} = 0.473$ vs run **[computed]**: depression 0.378 predicted vs 0.358 K measured (5.3%); v_{max} 9.18×10^{-3} vs $8.81 \times 10^{-3} \text{ m/s}$ (4.2%); edge 2.54×10^6 vs jet $2.36 \times 10^6 \text{ m}$ (−7.1%).

Consequences for the suite: (i) the $\text{Ro} \rightarrow 1$ anchor needs reduced ϵ_u , so the Tier-3 drag ladder is load-bearing rather than optional; (ii) at the $\sin^2$ forcing the circulation is $\sim 3\times$ stronger, so the drag share is $\sim 3\times$ smaller but still non-negligible at $v_d = 0$ (consistent with Zhang et al. 2025 Fig. 4’s max $\text{Ro} \sim 0.9$ there); (iii) the steady-state detector (window 30 d, 10^{-4}) fires while the jet still drifts $\sim 0.4\%$ per 60 d on the drag timescale, so budget-grade runs need a stricter threshold or a fixed post-detection extension.

2026-07-16, profile-level addendum to runs 1a–1b

The tables above are endpoint checks (the equatorial depression and v_{max} are, respectively, the $y = 0$ anchor of T and a single point of v). The suite standard is now the full-profile scorecard (Section 6), applied retroactively to both runs: final-30-day means, theory at the fitted cell-mean Ro of 0.479 (1a) and 0.473 (1b). Under this standardized window 1b’s fit reproduces the value quoted above exactly, while 1a’s shifts from 0.470 to 0.479, within run 1a’s stated $\sim 2\%$ drift allowance. Profile results **[computed]**:

run	field	max $ \Delta $ (cell)	at y [10^6 m]	% of scale	inner half
1a	u	10.4 m/s	2.52	34	2.3 m/s (7.5%)
1a	T	0.133 K	2.01	36	0.038 K (10%)
1a	v	4.0 mm/s	2.44	45	0.49 mm/s (5.5%)
1b	u	10.2 m/s	2.52	33	3.2 m/s (10%)
1b	T	0.152 K	2.01	40	0.042 K (11%)
1b	v	3.9 mm/s	2.44	43	0.48 mm/s (5.2%)

Scales: cell-edge $u_{\text{Ro}}(Y_{\text{Ro}})$, the equatorial depression, and v_{max} . Reading: the inner half of the cell agrees at the 5–11% level in all three fields, and the misfit concentrates at the terminus in all three, consistent with the diagnosed local $\text{Ro}(y)$ declining to ~ 0.2 at the jet: a single cell-mean Ro is furthest from the local value exactly there. The simulated subtropical warm lobe of $D(y)$ peaks at 0.17 K (1a) against the predicted 0.30 K and is smeared poleward across the edge (D stays $\sim +0.015$ K out to at least 4.5×10^6 m), suggesting the sharp-edge equal-area closure, not the quartic interior shape, is the limiting idealization. The u misfit sits at the theory’s terminus discontinuity (the jump to the RCE plateau). The ms’s linear-Ro variant targets exactly this regime and is testable against these same stored profiles (suite extension, not yet scheduled).